

PROJECT PLANNING PHASE

Product Backlog, Sprint Planning, User Stories & Story Points

Date: September 28, 2026	Team ID: SWTID-2026-9692
Project Name: AI WeatherWise API	Marks: 5 Marks

1. Product Backlog, Sprint Schedule & Estimation

Sprint	Epic	Story ID	User Story / Task Description	Points	Priority
Sprint-1	User Auth	USN-1	As a user, I can register for the application by entering name, email (<code>shree@gmail.com</code>), and password (<code>1234567</code>).	2	High
Sprint-1	User Auth	USN-2	As a user, I receive a JWT bearer token and created user ID upon successful registration.	1	High
Sprint-1	User Auth	USN-3	As a user, I can log into the application with valid email and password credentials.	1	High
Sprint-2	Weather API	USN-4	As a user, I can fetch live weather data for Chennai (<code>/api/weather/chennai</code>) showing temperature, humidity, and condition.	3	High
Sprint-2	AI Engine	USN-5	As a user, I can request an AI-generated weather summary (<code>/api/ai/weather-summary</code>) based on current weather data.	5	Medium
Sprint-2	AI Engine	USN-6	As a user, I can request tailored weather recommendations (<code>/api/ai/weather-recommendation</code>) for clothing and activities.	5	Medium
Sprint-3	Maintenance	USN-7	As a developer, I can update the Gemini API model configuration from <code>gemini-2.5-flash</code> to <code>gemini-3.8-flash</code> to resolve 404 endpoint errors.	3	Low

2. Project Tracker, Velocity & Burndown Chart

Sprint	Total Points	Duration	Start Date	Planned End Date	Points Done	Actual Release
Sprint-1	4	3 Days	20 Sep 2026	23 Sep 2026	4	23 Sep 2026
Sprint-2	13	4 Days	24 Sep 2026	28 Sep 2026	13	28 Sep 2026
Sprint-3	3	2 Days	29 Sep 2026	30 Sep 2026	0 (Planned)	30 Sep 2026

3. Velocity Calculation

Team Average Velocity (AV) per iteration unit (story points per day):

AV = Total Story Points Completed / Total Sprint Days

• Sprint-1 Velocity: 4 points / 3 days = 1.33 points/day

• Sprint-2 Velocity: 13 points / 4 days = 3.25 points/day

• Overall Average Team Velocity: 17 completed points / 7 total days = 2.43 story points per day

4. Burndown Chart & References

Burndown Chart Definition & Documentation: A burndown chart is a graphical representation of work left to do versus time in agile software development. It tracks overall progress toward completion over each iteration.

Key Documentation Links:

- [Atlassian Agile Project Management](#)
- [Atlassian Epics Tutorials](#)
- [Atlassian Sprints Tutorials](#)
- [Atlassian Agile Estimation](#)
- [Atlassian Burndown Charts Tutorials](#)